

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL ONE MODULE, AUTUMN/SPRING SEMESTER 2021-2022

Mathematics for Computer Scientists 1

Time allowed: ONE Hour and THIRTY Minutes

This is an open-book shorter-timed examination. Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

The full marks are 100. You MUST answer question 1 worthy of 25 marks, and answer only TWO questions from question 2 to question 4, each worthy of 33 marks. 9 marks are allocated for submission in $\LaTeX$

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

Question 1: Computational logic and sets.**[overall 25 marks]**

- a. Answer true or false for the following questions. p and q are Boolean variables. $\mathbf{S}$ is a set. $\sim \mathbf{S}$ is the complement of $\mathbf{S}$. $\emptyset$ is the empty set.
- i). $(p \Rightarrow q) \wedge (p \Leftarrow q) \equiv (p \equiv q)$.
 - ii). $\mathbf{S} = \emptyset, |2^{\mathbf{S}}| = 1$.
 - iii). $\sim \mathbf{S} \cap \mathbf{S} = \emptyset$.
 - iv). $\neg(p \vee q) \equiv \neg p \vee \neg q$.
 - v). $\neg(p \Rightarrow q) \equiv p \Leftarrow q$.

[10 Marks]

- b. Consider any set $\mathbf{A}$ written out as a list of ten elements, of which two of them are exact copies. For example, $\mathbf{A} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 9\}$. First, mathematically prove for any set $\mathbf{A}$ that the order in which the elements are listed out does not matter. Second, mathematically prove for any set $\mathbf{A}$, the individual elements are counted once so that $|\mathbf{A}| = 9$. Full marks using only basic mathematics and notation as given out in the lecture materials (e.g. set comprehension, propositional laws).

[15 Marks]**Question 2: Questions on Functions, Relations, and Matrices.****[overall 33 marks]**

- a. For each of the following statements, say whether it is **true** or **false**.
- i). There are in total $|A| \times |B|$ different relations from a finite set A to a finite set B .
 - ii). There are in total $|A|^{|B|}$ different total functions from a finite set A to a finite set B .
 - iii). Let $\mathcal{Y}$ be the set of real functions. Define the relation R from $\mathbb{R}$ to $\mathcal{Y}$ such that aRf if and only if $f(a) = 0$. Therefore, $\neg 1R(x^2 - 1)$.
 - iv). The "is a superset of" relation $\supseteq$ is the inverse of $\subseteq$.
 - v). Define f from $\mathbb{Z}^+$ to $\mathbb{Z}^+$, where $\mathbb{Z}^+$ denotes the set of positive integers, such that $f(n) = 3n^2 - 2$. Based on the definition, f is an injective function.

[5 Marks]

- b. Let f , g , and h be functions from $\mathbb{R}$ to $\mathbb{R}$, where $\mathbb{R}$ denotes the set of real numbers. Note the definitions of the operations $+$ and $\cdot$ are given in the lectures.
- i). Show that $(f + g) \circ h = f \circ h + g \circ h$.

[4 Marks]

- ii). The sum of two subsets A and B of $\mathbb{R}$ is defined by $A + B = \{c : \exists a \in A, b \in B, c = a + b\}$. Show that the $\text{range}(f + g) \subseteq \text{range}(f) + \text{range}(g)$, where $\text{range}(f)$ and $\text{range}(g)$ denote the range of function f and g , respectively.

[7 Marks]

- iii). Define functions f, g, h by $f(x) = -5x + 2$, $g(x) = 5x^2 - 7$, and $h(x) = (x - 3)/(x + 2)$, respectively. Determine whether each of them is a bijection. Explain your answer.

[3 Marks]

- c. SoftJobs is a company capable of producing a variety of software packages, ranging from data management software to video games. The company uses a database to keep track of which programmers are working on which projects for which clients, shown in Table 1.

Project	Client
Football Dunkfest	GameCo
Rx Tracker	MediComp
Skate Rats	GameCo

(a) Contracts

Programmer	Project
Nurul	Football Dunkfest
Ali	Rx Tracker
Chan	Football Dunkfest
Nanda	Skate Rats
Rajeswari	Skate Rats

(b) Work Assignments

Table 1: Database for SoftJobs

- i). Represent the relations depicted in Table 1a and 1b with binary matrices, respectively.

[8 Marks]

- ii). Compose the two matrices you obtain in item c.i) and determine for which client Nurul is working based on the result of composition.

[6 Marks]

Question 3: Questions on Numbers and Matrices.

[overall 33 marks]

- a. For each of the following statements, say whether it is **true** or **false**.

- Let a and m be integers with $m \geq 1$, then $a|a^m$.
- 2 is the only even prime number.
- $\forall x, y \in \mathbb{R}, \lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$.
- 35 hours after 2pm would be 1pm.
- $\gcd(111, 51) \equiv 3 \pmod{3}$

[5 Marks]

- b. Show that, if an integer $n > 1$ is composite, then n has a factor larger than 1 and less than or equal to $\sqrt{n}$.

[5 Marks]

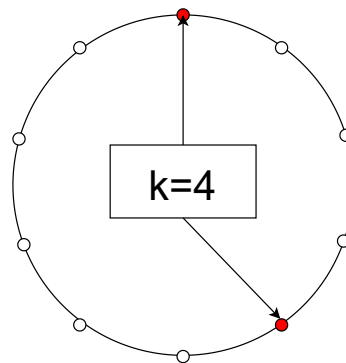


Figure 1: Cycle of $n = 10$ beads with skipping number $k = 4$

- c. Suppose n beads are to be arranged on a circular loop, and every k th bead must be the same colour. That is, for each bead b , both beads k positions away from b must be the same colour as b . Figure 1 shows a case where $n = 10$ and $k = 4$. In general, for a cycle of n beads and a skipping number k , what is the maximum number of different colours that could be used? Explain. (Hint: $nk = \text{lcm}(n, k) \cdot \text{gcd}(n, k)$)

[10 Marks]

- d. Given a permutation $p_1, p_2, \dots, p_n$ of the integers $1, 2, \dots, n$, the corresponding permutation matrix is the n -by- n matrix $P_n = [a_{i,j}]$ such that

$$a_{i,j} = \begin{cases} 1 & \text{if } p_i = j \\ 0 & \text{otherwise} \end{cases}$$

- i. Determine the permutation matrix according to the permutation 3, 2, 5, 4, 1.

[4 Marks]

- ii. Show that for any n , $P_n \cdot P_n^T = I_n$ where I_n is an n -by- n identity matrix.

[9 Marks]

Question 4: The following questions are pertaining to Counting and Probability.

[overall 33 marks]

- a. For each of the following statements, say whether it is **true** or **false**.
- There are in total $10 \cdot P(9, 4)$ possible different number plates consisting of 6 digits (0 to 9) with exactly 2 of the same digit.
 - Three standard (6-sided) dice are rolled. One is red, one is blue, and one is green. If the particular number showing on each coloured die is important, then there are in total 6^3 possible different outcomes.
 - A bag contains 10 black and 12 white balls and each ball is considered distinguishable from each other. There are in total $C(10, 5)C(12, 3) \cdot C(10, 6)C(12, 2)$ different ways to select 8 balls so that exactly 5 or 6 are black.
 - Two 6-sided fair dice are rolled. The probability that at least one of the dice is 6, is $\frac{1}{3}$.

- v. Tossing a fair coin and getting a head is mutually exclusive from rolling a 6-sided die and getting a 5.

[10 Marks]

- b. A necklace is to be formed by threading 8 beads on a circle of string. The beads are taken from a bag containing 14 different beads. How many options are there for a necklace if the knot holding the string together is not visible? Explain your answer.

[5 Marks]

- c. An insurance company issues car insurance, homeowner insurance, and life insurance. Of the claims the company pays, 50% are car, 40% are homeowner, and 10% are life. The company is primarily concerned with claims over RM50,000 that it pays. History has shown that 10% of car claims, 30% of homeowner claims, and 90% of life insurance claims exceed RM50,000.

- i. What percentage of claims paid by this company exceed RM50,000?

[12 Marks]

- ii. Given that this company pays a claim over RM50,000, what is the probability that it is a homeowner claim?

[6 Marks]